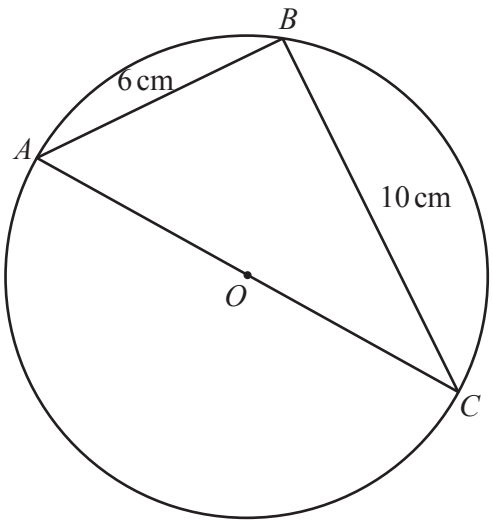
0580_s21_qp_31_Q8

8 **(a)**



NOT TO SCALE

A, B and C lie on a circle, centre O, diameter AC.

(i) Complete this statement.

Angle ABC is 90° because ..... [1]

(ii) Work out the area of triangle ABC.

..... cm^2 [2]

(iii) Work out AC.

$AC =$ ..... cm [2]

(b) Make r the subject of the formula $A = \pi r^2$.

$r =$ ..... [2]

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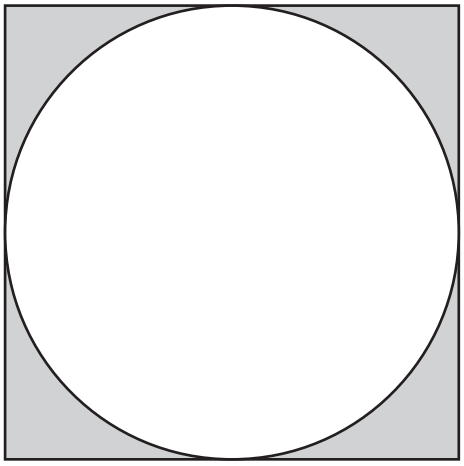
(c)



NOT TO SCALE

The diagram shows a circle inside a square.
The circle touches the four sides of the square.
The area of the square is $81\,\text{cm}^2$.

Calculate the shaded area.

..... cm^2 [4]